

EE2 Circuits & Systems

Lab 0 – Getting Ready with Lab-in-a-Box

Peter Cheung, 11 October 2022

Introduction

This document outlines the preparation you need to make before you start the Laboratory Sessions for this module. In a sense, this is Lab Session 0.

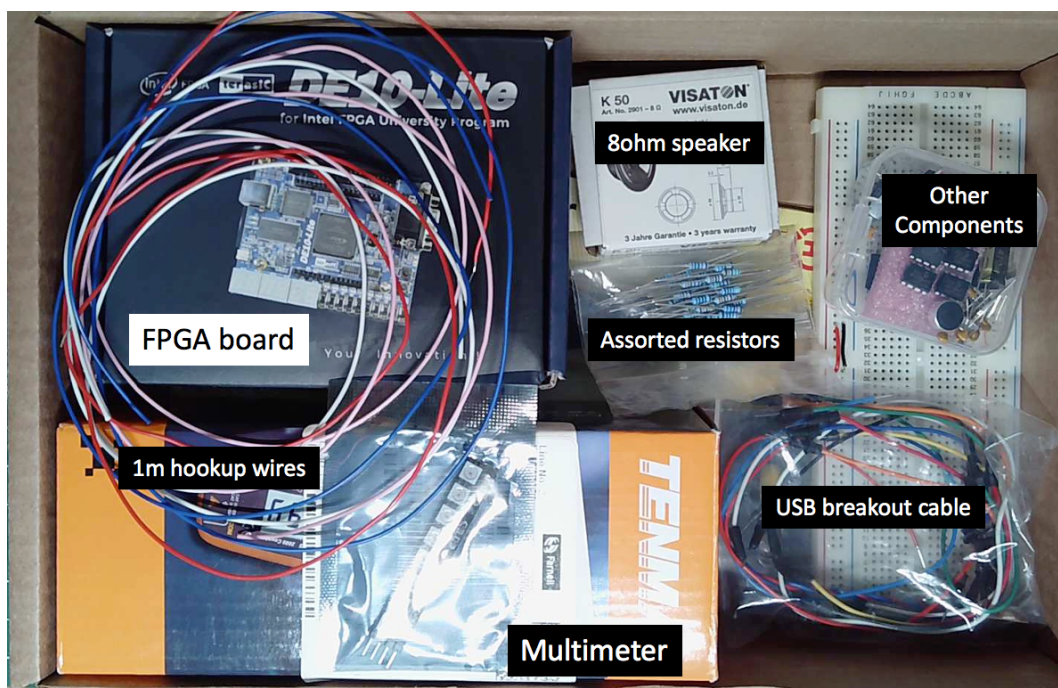
Opening the Lab-in-a-Box

Since we prepared laboratory delivery under COVID, we now provide each student with their own kit of you will take possession of a Lab-in-a-Box kit. You can pick up your named box from Level 1 Electronics Lab (from May or Vic). Appendix A shows the packing list for the laboratory kit.

The parcel contains two boxes inside. It consists of two measurement instruments: a USB-based oscilloscope (PicoScope) with two scope probes and a BNC cable, and a multimeter and a pair of measurement probes. **These are on loan to you for use throughout your 2nd year and you will need them for laboratory experiments in all three terms.** Therefore, you MUST keep all the packing materials to facilitate their return at the end of the 2nd year.

There is also a second box containing an FPGA board (DE10-Lite) with a prototype plug-in card. This is also on loan to you specifically for the Circuits and Systems module. Put this box to one side for the moment because you will not be using it until Lab 3. You will be instructed to return this to us at the end of the Autumn term because they will be needed for 2nd year EIE students in the Spring term.

The second box also contains other components: a full-size breadboard, a plastic box containing electronic components for you to build circuits, a wire stripping tool, 1m long hookup wires in four different colours, etc. Check that the box has what you should expect as shown in the photo below.



Sorting and Storing electronic components

The contents of the Lab-in-a-Box parcel will have to serve you throughout the Autumn term and beyond. Therefore, this next step is vital if you want to avoid missing components later in the term. Some components are very small and easily lost. Missing a humble resistor may prevent you from completing some of the experiments later. Of course, you can pick up spare components from Level 1 Lab, but only when the Lab is open.

1. Prepare two cardboards or stiff sheets of paper. Write onto the cards the component labels as shown in Appendix B. If you have a printer, print these pages and stick onto stiff cards.
2. Sort out the components by type and value. You may need to use the multimeter to measure values of the resistances and capacitances and make sure that the colour code and capacitor markings match those of the measured values. You can find the User's manual for the Tenma 72-2985 multimeter by clicking on the QR code.
3. Use sticky tape and tape down each component type under the corresponding label.
4. Be sure that you return all components back to your **cardboard store** after use. Otherwise, you will not be able to find the component later on during the term. The discipline of storing components back in the cardboard store will save you the trouble of having to find replacement component in the Lab later.



Preparing the breadboard

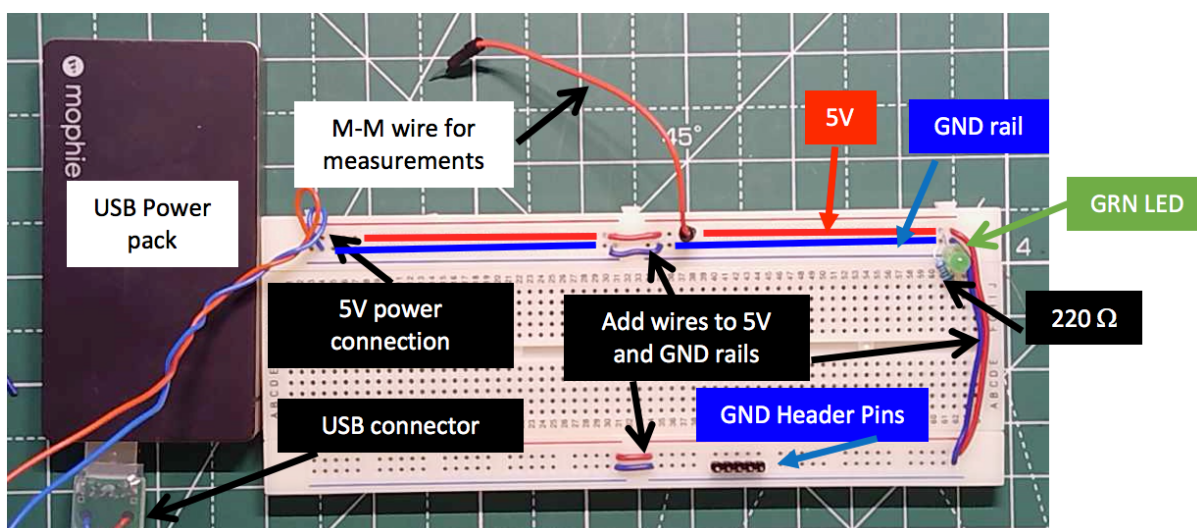
You have used breadboards in your first year already and should know how to build circuits with them. You will be building circuits on the breadboard for Lab 1 and 2.

The breadboard has four regions. The middle part is for your circuits, separated into upper and lower regions, which are not connected together. In either regions, the vertical contacts (in rows ABCDE, and FGHIJ) are shorted together. The upper and lower perimeters are the power rail regions. The horizontal rows are connected from left to right across the entire breadboard. They are marked with RED and BLUE lines. This colour code matches the power USB connector provided.

You **MUST** always use the RED row as 5V rail, and the BLUE row as GND rail.

Connect the GND and 5V rails as shown in the diagram. Connect the USB power cable to the breadboard and to a USB 5V source. You can either use the USB output from your laptop or, even better, use a USB power pack, or charger. Note that the biggest danger in destroying your chips is if you connect the power and ground wire the wrong way round. So take care! Add the GREEN LED in series with a 220 ohm resistor to monitor the 5V rail.

Use the multimeter, check the voltage between RED and BLUE is indeed 5V. Insert the 5-way header pin to lower ground rail. This serves as a convenient anchor point for Earth Clip for the scope probes.



Setting up the PicoScope

Step 1: Download from the Pico Technology website the PicoScope software for your computing. The model you have is PicoScope 2204A (i.e. 2000 Series). Install the software on your computer. You can reach the site in three ways: 1) click on the QR code on the right; 2) use your photo camera on the QR code; 3) via a link in the Circuits and Systems course webpage.



Step 2: Download the User's Guide and other relevant documents for PicoScope 6 from this site shown on the right.

Step 3: Connect the PicoScope to your laptop using the supplied USB cable, and attach the two scope probes provided to Channels A and B of the scope.

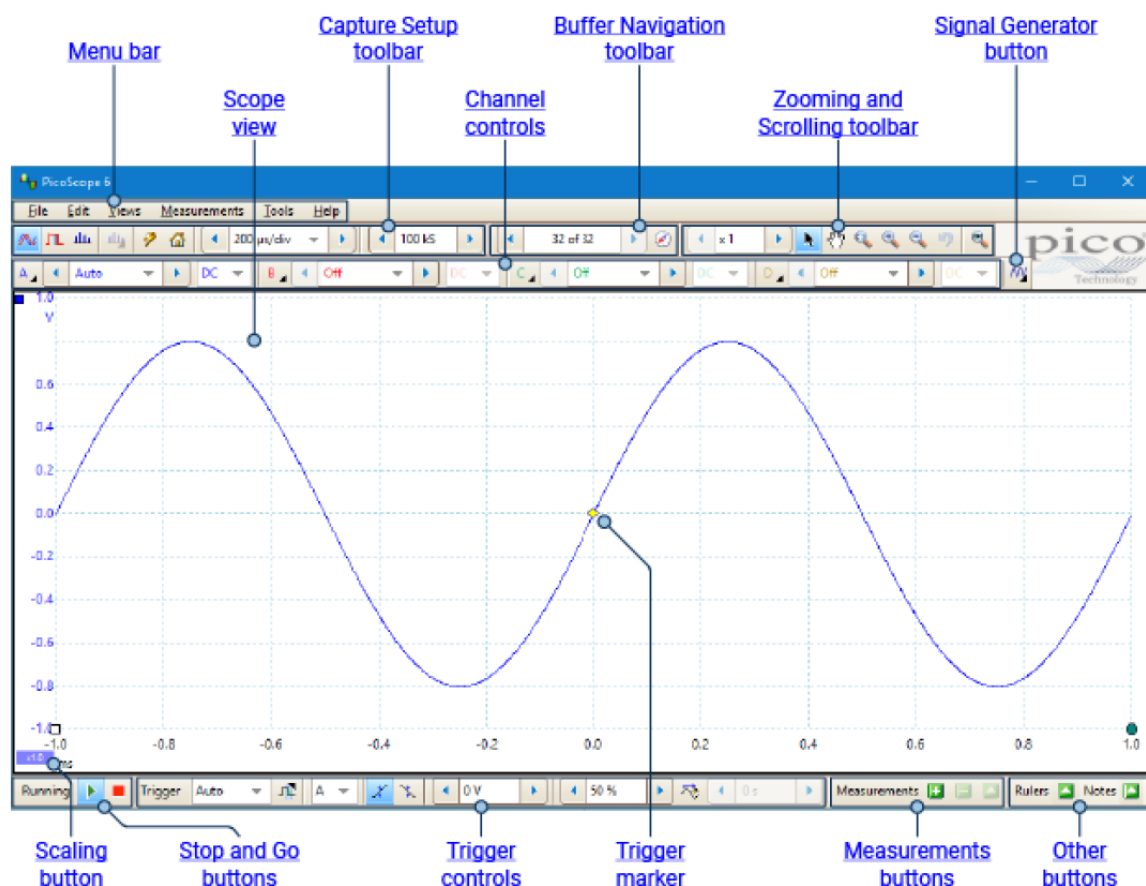
Step 4: Connect the BNC to clips to the Arbitrary Waveform Generator (AWG) output of the PicoScope.

You are now ready to use the PicoScope.

Learning to use the PicoScope and the waveform generator

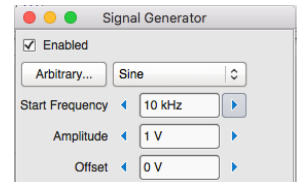
Using the PicoScope is slightly different from using laboratory oscilloscope in the EE Lab during your first year. It is therefore worthwhile for you to spend a bit of time familiarizing yourself with this new piece of equipment. Furthermore, the oscilloscope also includes a waveform generator which you will use as a signal source to your circuits.

Step 1: Launch the PicoScope 6 software on your laptop. You will see a window similar to that shown here. Note that the user interface on a Mac would be slightly different.



Step 2: Connect both scope probe inputs to the signal generator output and clip all the earth crocodile clips together.

Step 3: Generate a 10kHz sinewave with an amplitude of 1V (i.e. 2V pk-pk) by: 1) turn on the waveform generator with the AWG button ; 2) select the appropriate values in the popup form.



Step 4: Select on “Auto” Trigger and “Auto” scaling for Channel A. You should see a steady sinewave in blue colour being displayed. Make sure that both the amplitude and the frequency of the sinewave signal is as expected.

Step 5: Adjust the trigger level and trigger slopes and see how the displayed waveform changes.

Step 6: Turn off Channel A and turn on Channel B. Then turn on both channels. Explain what you see and why. Now change the various parameters such as voltage scaling from Auto to $\pm 5V$, or the time base (i.e. X-axis scaling).

Explore yourself

Explore the limitation of the signal generator by discovering answers to the following questions.

1. What is the maximum frequency of a sinewave that can be generated?
2. What is the maximum amplitude of a sinewave that can be generated?
3. Do the answers to 1) and 2) apply to square wave and triangular signals?
4. Generate a sinewave at 10kHz that goes from 0V to 2V (i.e. amplitude 1V and offset 1V). Is that possible? Can you adjust to offset to 1.2V (i.e. sinewave from 0.2V to 2.2V)?
5. A 3.3V digital signal goes from 0V to 3.3V. Generate a digital signal at 10kHz. Is it possible?

Appendix A
EE2 Lab-in-a-Box – List of Contents

Item	Qty. per kit
Boxed Items	
Picoscope 2204A with probes	1
DE10-Lite FPGA board	1
Multimeter Tenma 72-2985	1

Modules, wire, tools	
Prototype shield (inside DE10 FPGA box)	1
Component box (plastic)	1
Full-length breadboard	1
1m x 4 solid core wire hookup wires	4
BNC to clip lead (inside PicoScope box)	1
Wire stripper	1
Small screwdriver	1

C&S Component kit	
LM386 Audio Amp	1
MCP6002 Dual opamp	3
MCP4921 12-bit DAC	1
MCP3201 12-bit ADC	1
AP430i 2.5V reference	1
3.5mm audio jack breakout board	1
Speaker 8 ohm	1
Microphone - condensed	1
5V power connector to USB	1
Green, Blue & White LEDs	3
10k variable resistor, Alp	1
10 ohm 1W	1
Breadboard hookup wire (bundled)	16
4 or 5-way inline header pins	1

Resistors & Capacitors (from Lab)	
10k	6
100k, 200k	4
100, 220, 470, 1k, 1.5k, 2.2k, 3.3k, 4.7k, 6.8k	2 each
15k, 22k, 33k, 47k, 68k	2 each
150k, 220k, 330k, 470k, 560k, 1M	2 each
10nF, 33nF, 47nF, 100nF, 1uF	4 each
10uF electrolytic (6Vdc minimum)	3
470uF electrolytic	1
1N914 or equivalent small signal diode	2

Appendix B – Component Store Sheets

10 Ω	100 Ω	220 Ω	470 Ω
1k Ω	1.5k Ω	2.2k Ω	3.3k Ω
4.7k Ω	6.8k Ω	10k Ω	15k Ω
22k Ω	33k Ω	47k Ω	68k Ω
100k Ω	150k Ω	200k Ω	220k Ω

330k Ω	470k Ω	560k Ω	1M Ω
10nF	33nF	47nF	100nF
1uF	10uF	470uF	
IN914	AP430i	MCP6002 Op-amp	
LM386 Audio Amp	MCP4921 DAC	MCP3201 ADC	